



Contents lists available at ScienceDirect

Journal of Colloid and Interface Science

journal homepage: www.elsevier.com/locate/jcis

Regular Article

Engineered heat dissipation and current distribution boron nitride-graphene layer coated on polypropylene separator for high performance lithium metal battery

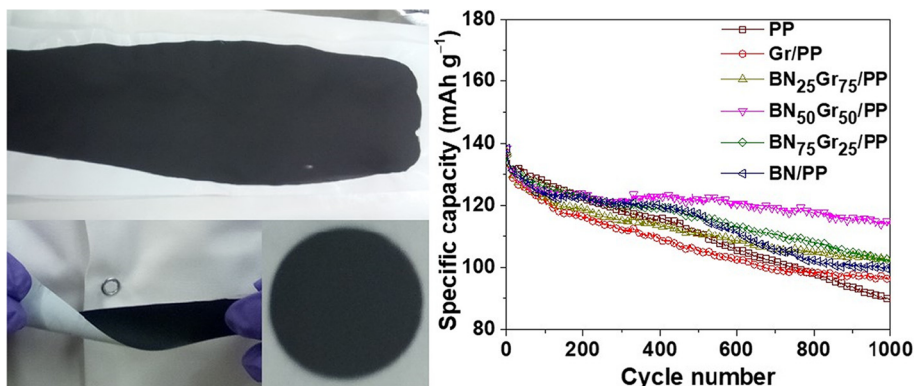
Jassiel R. Rodriguez ^{a,*}, Patrick J. Kim ^a, Kyungho Kim ^b, Zhimin Qi ^b, Haiyan Wang ^b, Vilas G. Pol ^a^a Davidson School of Chemical Engineering, Purdue University, West Lafayette, IN 47907, USA^b School of Materials Engineering, Purdue University, West Lafayette, IN 47907, USA

HIGHLIGHTS

- A BN_xGr_y layer on one side of PP separator suppresses the sharp Li dendrite growth.
- The bilayer BN_xGr_y/PP separator improves the half- and full-cell performances.
- Full cells with modified BN₅₀Gr₅₀/PP separator deliver 114 mAh g⁻¹ at 1 C after 1000 cycles.

GRAPHICAL ABSTRACT

The bilayer BN_xGr_y/PP separator deliver a superior performance as result of an effective synergistic effect between physico-chemical properties of Gr and BN.



ARTICLE INFO

Article history:

Received 3 July 2020

Revised 22 August 2020

Accepted 1 September 2020

Available online 15 September 2020

Keywords:

Bilayer BN_xGr_y/PP separator

Suppression of dendritic Li

Superior performance

Synergistic effect

Li metal battery

ABSTRACT

Li metal as a battery anode has been intensively studied because of its high gravimetric capacity (3860 mAh g⁻¹), a low standard electrode potential (−3.04 vs. SHE), a reasonable electronic conductivity and low density. However, lithium metal suffers from a continuous Li dendrite growth upon charge-discharge cycling, delivering a poor coulombic efficiency and consequently its early failure. Here, engineered bilayer separators demonstrate that a boron nitride-graphene (BN_xGr_y) layer coated on one side of polypropylene (PP) membrane remarkably reduces the polarization and impedance, and significantly improve the performance and stability of Li/Cu half-cells. Moreover, Li/LiFePO₄ full cell with the modified BN₅₀Gr₅₀/PP separator presents a remarkably stable 1000 charge-discharge cycles with a specific capacity of 114 mAh g⁻¹ at 1C-rate. The superiority of the modified separator is originated from an effective synergistic effect between physico-chemical properties of Gr (reducing local current density) and BN (dissipating local heat) and its enhanced structural and mechanical stability.

© 2020 Elsevier Inc. All rights reserved.

* Corresponding author.

E-mail address: rodri571@purdue.edu (J.R. Rodriguez).

1. Introduction

Energy storage devices such as lithium ion batteries (LIBs) have demonstrated a remarkable versatility to adapt to the continuous technological advances in a wide range of portable, electric vehicle and large-scale grid applications [1–3]. LIBs can store and supply electricity at a reasonable cost [3], as well as deliver high power and energy density with stable performance upon cycling [4]. However, LIBs have almost reached their theoretical limit because the commercial graphite anode stores a moderate capacity $\sim 330 \text{ mAh g}^{-1}$ at an applied current of 100 mA g^{-1} [5], which is not enough to satisfy the energy storage needs of the modern customers [6]. This issue has encouraged researchers to develop high capacity Li-host anode materials [7]. Among all the candidates, Li metal anode is the best choice because of its high capacity (3860 mAh g^{-1}), low standard reduction potential (-3.04 vs SHE) and lightness (0.59 g cm^{-3}) [8–10]. This is intrinsically related to increase the power and energy density of a secondary battery [11–13].

During the past decades, researchers have paid significant attention to stabilize the Li metal anode [14,15] to develop commercial Li metal batteries with better cycle performance, capacity and safety aspects [16,17]. However, this has not been achieved until now, because of the high instability of Li metal under cycling conditions [18]. Some approaches have been implemented to extend the cycle life and coulombic efficiency of Li metal batteries by (i) confining the metallic particles of Li in a 3D network [19–22], (ii) modifying the electrolyte by adding additives [23–25], (iii) coating a passivation layer on the anode [26–29], (iv) designing new separators with multifunctional features [30–34] and others [18,35].

The modification of a polypropylene (PP) membrane is one of the most versatile and feasible ways to enhance the coulombic efficiency and cycling performance of Li metal batteries [34]. In which, the modification of PP separator with functional materials can add some of the beneficial effects (such as mechanical strength and thermal conductivity) to the bare PP separator, which efficiently suppresses the Li dendrites and stabilizes the redox reactions. Few layered graphene presents a thermal conductivity of $1300 \text{ W m}^{-1} \text{ K}^{-1}$ and a Young modulus of 20 GPa [36], meanwhile thin BN flakes present a thermal conductivity of $750 \text{ W m}^{-1} \text{ K}^{-1}$ with a Young modulus of 80 GPa and similar hardness of $\sim 90 \text{ GPa}$ [37,38]. On the other hand, layered Gr-based compounds have promoted stable Li deposition in Li/Cu half-cell or Li/LiFePO₄ full cell configurations by reducing the local current density highly associated with Li dendrite formation [39,40]. Similarly, few layer BN nanosheets have increased the thermal conductivity of the PP separator, leading to a homogeneous Li plating/stripping in Li/Cu half-cell and thus suppression of sharp Li dendrite formation [41].

In this study, engineered bilayer separators with a boron nitride/graphene layer were designed to enhance the electrochemical performance of Li metal anodes. It is clearly confirmed by the cycle results of Li/Cu half-cell and Li/LiFePO₄ (Li/LFP) full cell that the BN/Gr layer decreased the local current density during the Li plating/stripping processes and enhanced the thermal conductivity and mechanical stability of conventional PP separators. In addition, it was found that the mass ratio of BN to Gr (50:50, by mass) delivered the best electrochemical performance and stability. When the designed separator was used, the cycle stability of Li/Cu half-cells improved almost two times in comparison with the Li/Cu cell using a PP separator. Moreover, all the modified separators retained higher capacity in the Li/LFP full cell after 1000 cycles than the pristine membrane, clearly indicating that the addition of an extra layer of Gr, BN or a mixture of them can improve the cycle performance of Li metal batteries.

2. Experimental part

2.1. Material preparation

Boron nitride ultrafine powder (99.0% BN with particle size of 70 nm and surface area of $20 \text{ m}^2 \text{ g}^{-1}$ from Graphene Supermarket), Graphene nanoplatelets (high purity GNPs, surface area of $300 \text{ m}^2 \text{ g}^{-1}$ from Sigma-Aldrich), *N*-methyl-2-pyrrolidone (99.5% C₅H₉NO from Sigma-Aldrich) and polyvinylidene fluoride (Kynar® PVdF-HV-900 from) was used as-received from vendor.

2.1.1. BN_xGr_y powder preparation

The set amount of BN powder and Gr nanoplatelets were grinded by hand in mortar and pastel and introduced into an Ar-filled glovebox. The mixture was put into a stainless-steel container of 50 mL with a stainless-steel ball with a diameter of 1 cm . Subsequently, the mixture was ball milled for 6 h at 600 rpm . Then, the final powder was collected and stored inside the Ar-filled glovebox. The same procedure was repeated for BN_x-Gr_y preparation with different mass ratio ($x:y = 0:100, 25:75, 50:50, 75:25$ and $100:0$).

2.1.2. Modified BN_xGr_y/PP separator preparation

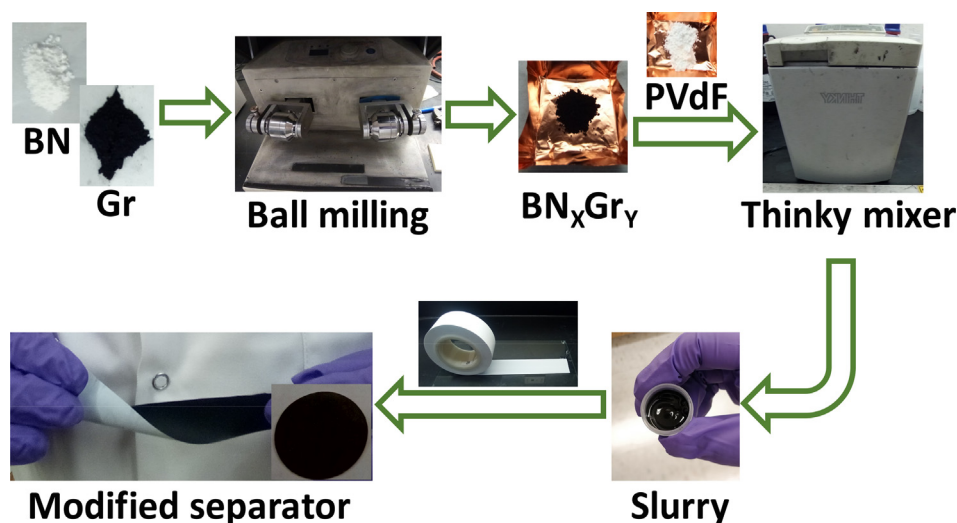
Scheme 1 shows the procedure to prepare the modified bilayer BN_xGr_y/PP separator. First, 40 mg of the prepared BN_xGr_y powder was mixed for 20 min with 10 mg of PVdF in NMP solution using a Thinky mixer. Then, the resultant slurry was casted on Celgard™ 2500 polypropylene membrane by doctor blade. Next, the as-prepared separator was dried at 50°C overnight under vacuum to remove the organic solvent. After drying, the modified bilayer BN_xGr_y/PP membrane was prepared as disk separators with a diameter of 16 mm for 2032 coin-cells (see Scheme 1).

2.2. Material characterization

The crystal structure of samples was acquired using a Rigaku Smart-Lab high-resolution diffractometer for powders under Bragg-Brentano geometry. Raman spectrum for BN and Gr structural characterization were collected using a Thermo Scientific™ DXR™3 Raman microscope equipped with a red laser (wavelength 633 nm). N₂ physisorption was carried out to determine the textural properties of the samples using a Quantachrome Nova 2200e surface area and pore size analyser. The topographic and microstructural analysis of the samples were examined using a JEOL NeoScope JCM6000 Benchtop microscope at 45 kV and a FEI Talos microscope at 200 kV . Additionally, the elemental information of the powders was authenticated via energy-dispersive X-ray spectroscopy (EDS).

2.3. Electrochemical evaluation

The electrochemical properties of the modified BN/Gr separators were studied through galvanostatic (dis)charge cycling and potentiostatic electrochemical impedance spectroscopy (EIS). The Li dendrite suppression properties of the modified separators was evaluated in Li/Cu half-cell, using Li and Cu disks as counter and working electrodes, respectively, and 1.0 M LiPF_6 in EC:DEC as an electrolyte. Also, Li/LiFePO₄ full cells were evaluated to analyze the effect of BN/Gr layer on its stability and performance. The cathode was prepared by mixing LiFePO₄, super P carbon, and PVDF with a mass ratio of $8:1:1$ in NMP solvent. The resultant slurry was casted on an Al foil and dried at 80°C overnight under vacuum. The film was used to obtain cathodic electrodes with a diameter of 12 mm to assemble full cells in coin-type cells 2032, using Li



Scheme 1. Procedure followed to prepare the bilayer BN_xGr_y /PP separator.

foil as anode, the bilayer membranes as separator, and 1.0 M $LiPF_6$ in EC:DEC as an electrolyte.

3. Results and discussion

After the preparation of BN_xGr_y slurries, these were used to cover one side of the PP membrane using doctor blade technique, as shown in Fig. 1a. The resultant bilayer separator can be bent and twisted without any damage (Fig. 1b, e). During the casting process the pristine separator was attached with adhesive tape to the Al foil and stretched gently to cast a uniform porous BN_xGr_y layer with a thickness of $\sim 10\ \mu m$ (Fig. 1d, f). Through this process, a homogeneous BN_xGr_y layer was successfully deposited on the side of PP membrane with similar features (Fig. 1c, f). Also, the wetting test (Fig. S1) reveals that the bilayer BN_xGr_y /PP separators present a fast and effective soaking of electrolyte, indicating better wettability in comparison with the bare PP separator. Additionally,

the electrolyte uptake test reveals that the bilayer $BN_{50}Gr_{50}$ /PP separators stores 2.3 times more electrolyte than the bare PP separator. Thus, the BN_xGr_y film on the PP membrane allows an effective penetration and high uptake percentage of electrolyte into the bilayer separators.

After the milling process, powder X-ray diffraction and Raman spectroscopy was performed to investigate the change of structure after the high energy ball mixing process, see Fig. 2. As the hexagonal BN (*h*-BN) and Gr crystal structure are similar as graphite [42,43], their diffraction peaks appear at similar 2θ positions, according to the reported ICDD PDF file 034-0421 and 001-0640. *h*-BN diffraction peaks appear at 2θ values of 26.9° , 41.8° , 44° , 50.3° and 55.2° , meanwhile Gr diffraction peaks appear at 26.3° , 42.6° , 44.8° , 50.3° and 54.2° , corresponding to the same indexed plane (002), (100), (101), (102) and (004), respectively. Consequently, the (002) peak of Gr is slightly shifted at high values of 2θ as BN concentration increases in the samples, because the *c* lattice constant from *h*-BN crystal ($6.6\ \text{\AA}$) is smaller than that from

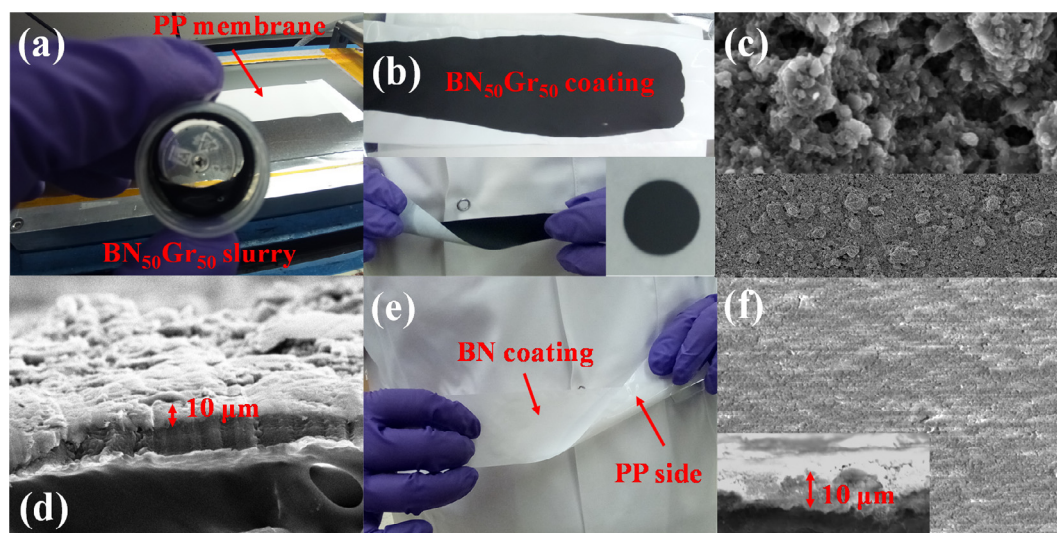


Fig. 1. (a) Images of $BN_{50}Gr_{50}$ -based slurry before casting it on the PP membrane. (b) Coated and twisted bilayer $BN_{50}Gr_{50}$ /PP separator after drying process. (c) Frontal and (d) cross-section SEM images of the bilayer $BN_{50}Gr_{50}$ /PP separator. (e) Twisted bilayer BN/PP separator. (f) Frontal and cross-section SEM images of the bilayer BN/PP membrane.

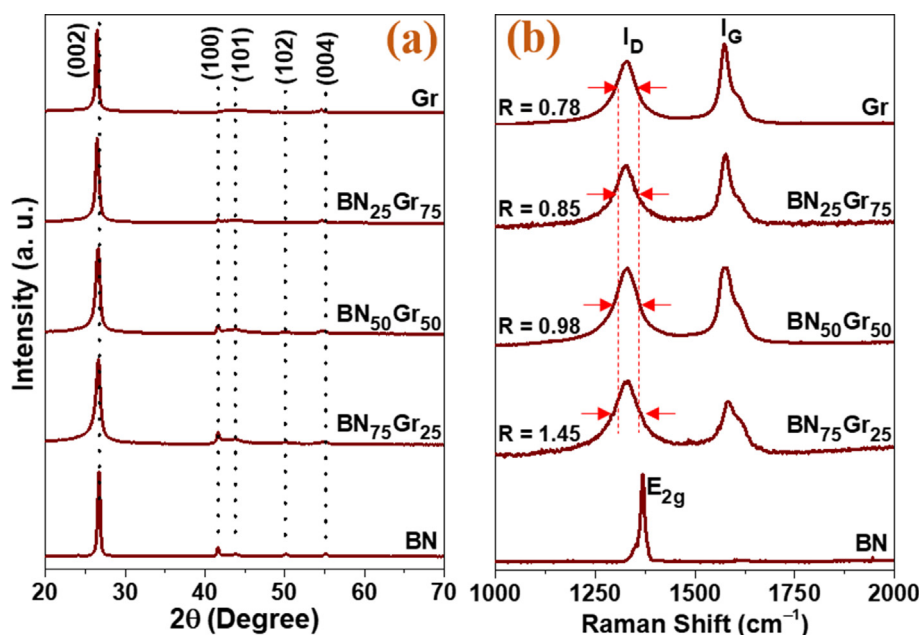


Fig. 2. Structural characterization of the BN_xGr_y powders: (a) X-ray diffraction patterns; and (b) Raman spectra.

graphite (6.8 Å). The high intensity of the indexed plane (002) in all the samples indicates an extent of interplanar ordering and suggests that a homogeneous dispersion of BN-Gr nanosheets was obtained after the ball milling process.

Raman spectra present two characteristic peaks at 1328 and 1573 cm^{-1} , which correspond to the D- and G-bands, respectively (see Fig. 2b). The D-band is ascribed to the presence of disorder or vacancies in the few layer graphene nanoplatelets [44], while the G-band is associated to sp^2 stretches from hexagonal C—C bonds in the crystal structure [44]. The amorphousness of the samples increases with the amount of BN according to R values [44], which correspond to the ratio of the D- and G-band areas. Also, the full width of half maximum of the I_D peak grows in parallel with the amount of BN because its counter-phase active $\text{E}_{2\text{g}}$ vibration mode appears at 1366 cm^{-1} [45], and then gives some contribution to the intensity and broadness of the I_D peak. These structural changes observed among the different Raman spectra could promote a significant impact on the cycling performance of Li metal cell with the modified bilayer $\text{BN}_x\text{Gr}_y/\text{PP}$ separators.

Topographical images of the raw BN nanoparticles and Gr nanoplatelets is shown in Fig. S2. In which, it is clearly observed that after ball milling both materials to produce the sample $\text{BN}_{50}\text{Gr}_{50}$, the resultant powder displays a flake-like topography similar to the Gr nanoplatelet, suggesting the formation of a homogeneous dispersion of the BN nanoparticles on the Gr nanoplatelets. Fig. 3a shows the low-resolution TEM micrographs of the raw BN nanopowder, which exhibits a rod-like morphology with a particle size >200 nm. Also, a HR-TEM image of the BN powder reveals that the thick nanoparticles (thickness >20 nm) are formed by multi-layer BN with an interplanar distance of 0.33 nm that correspond to the indexed plane (002) of the *h*-BN (Fig. 3a and S2c). In contrast, Fig. 3b–f present the TEM micrographs of the as-prepared BN_xGr_y samples, which clearly show that all the ball milled powders are conformed by thin nanosheets, suggesting that the one-pot and high energy process could transform the morphology of the big BN particles to thin BN nanosheets mixed with Gr nanoplatelets.

In Fig. 4a, a low-resolution TEM micrograph from sample $\text{BN}_{50}\text{Gr}_{50}$ reveals the presence of some thin nanosheets that shows a lot

of defects. Through a high-resolution TEM analysis (Fig. 4b, c), it is demonstrated that after ball milling the sample $\text{BN}_{50}\text{Gr}_{50}$, the BN particles acquired a nanosheet shape with few layers of thickness ~5 nm, with interplanar distances of 0.33 nm and 0.22 nm that correspond to the indexed planes (002) and (10 $\bar{1}$ 0) of the *h*-BN. The selected area electron diffraction (SAED) pattern from sample $\text{BN}_{50}\text{Gr}_{50}$ (Fig. 4d) confirms the characteristic laminar structure of the *h*-BN and Gr nanoplatelets, which are graphite-like structures with analogous XRD patterns. In which, it identifies the diffraction rings that corresponds to the family of indexed planes {002}, {100}, {101}, {102} and {002}. Furthermore, the chemical composition of the sample was investigated through EDX mapping to confirm the presence of boron (red), nitrogen (purple), and carbon (green), as given in Fig. 4(e–h). This analysis reveals a high dispersion and homogeneous distribution of the BN nanosheets on the Gr nanoplatelets in the $\text{BN}_{50}\text{Gr}_{50}$ powder.

In order to evaluate the electrochemical properties of Li metal cell with modified $\text{BN}_x\text{Gr}_y/\text{PP}$ separators, these were studied in the Li/Cu half-cell configuration with a capacity of 1000 $\mu\text{Ah cm}^{-2}$ at 0.5 mA cm^{-2} for Li plating, as given in Fig. 5. All the half cells exhibit a similar initial coulombic efficiency of around 90% with a stable plating and stripping performance at least until cycle 40th. The cycling performance of Li/Cu half-cell with a pristine PP membrane presents a sudden decay after 40 cycles (see Fig. 5g), exhibiting a huge polarization of 275 mV after 100 cycles (Fig. 5a) and a poor electrochemical stability and coulombic efficiency of 24.5%, as a result of its poor thermal and mechanical properties in comparison with the modified BN_xGr_y separators (Fig. 5b–f). The BN layer improves in 10 cycles the stability of the PP membrane with a cell polarization of 104 mV after 100 cycles (Fig. 5f, g), meanwhile the bilayer Gr/PP separator shows a stable performance during 80 cycles (Fig. 5b, g), suffering a polarization of 131 mV after 100 cycles with a coulombic efficiency of 48%. All the modified separators with Gr content present an intercalation reaction (gray dotted circle in Fig. 5b–e) that clearly suppress the initial polarization of the Li/Cu half-cell (see Table S1). In addition, this process can minimize the polarization of the half-cell during the plating and stripping cycles, improving the stability and coulombic efficiency of the Li/Cu half-cells. The modified $\text{BN}_{50}\text{Gr}_{50}$

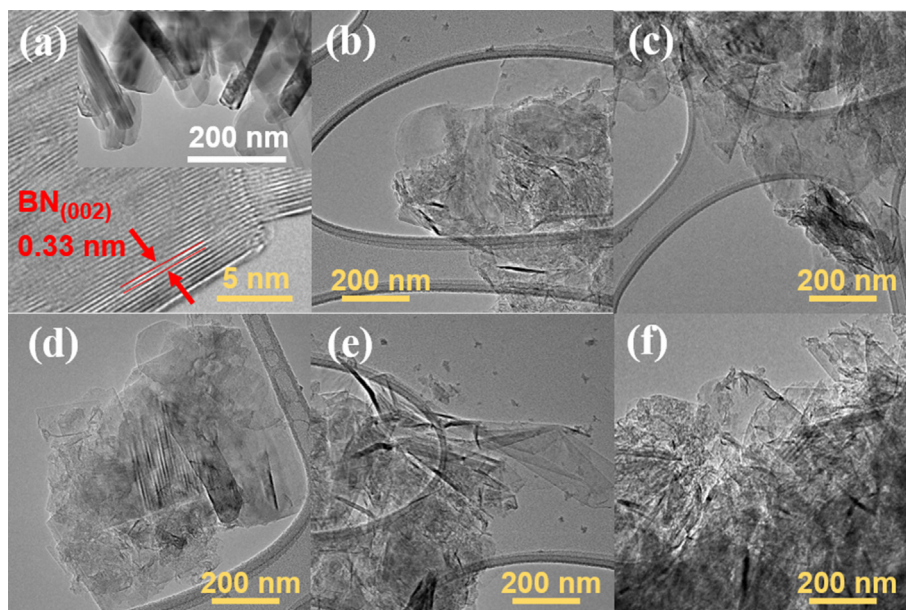


Fig. 3. (a) High- and low-resolution TEM micrographs of the pure BN powder. Low-resolution TEM micrographs of the samples (b, c) $\text{BN}_{75}\text{Gr}_{25}$, (d) $\text{BN}_{50}\text{Gr}_{50}$, and (e, f) $\text{BN}_{25}\text{Gr}_{75}$.

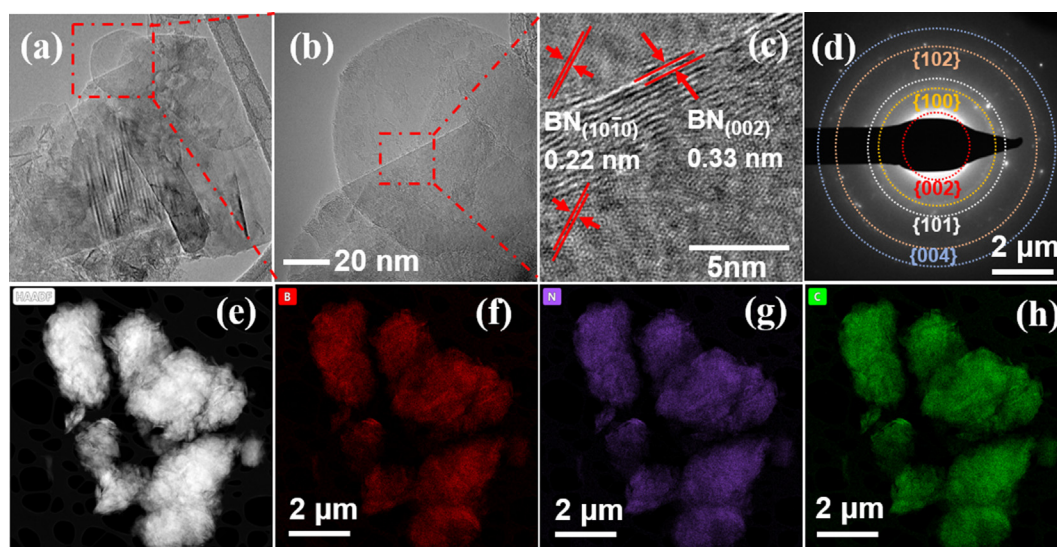


Fig. 4. Structural and chemical characterization of $\text{BN}_{50}\text{Gr}_{50}$ powder: (a, b) low- and (c) high-resolution TEM micrographs; (d) general TEM image with its selected-area electron diffraction pattern; (e)–(h) EDS analysis of some $\text{BN}_{50}\text{Gr}_{50}$ nanoparticles, boron (B, red), nitrogen (N, purple), and carbon (C, green). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Gr_{50}/PP separator displays the best plating/stripping behavior with the lowest half-cell polarization of about 88 mV after 100 cycles (Fig. 5d) with stable cycle performance and a high coulombic efficiency of 83.5% (Fig. 5g).

Also, the Li plating/stripping properties of the bilayer $\text{BN}_x\text{Gr}_y/\text{PP}$ separators were analyzed via a rate capability study at different values of current (Fig. 5h). In which, it is observed that the modification of the pristine PP separator with a layer of BN_xGr_y increases its coulombic efficiency approximately 3%, from 91% to 94%, at low current (0.1 mA cm^{-2}). Also, at high current, 1.0 mA cm^{-2} , the coulombic efficiencies of the modified $\text{BN}_x\text{Gr}_y/\text{PP}$ separators are around 7% higher than that one delivered by the bare PP separator. These could be due to the (de)lithiation reactions that occur in the

layer with Gr, because these reactions mitigate the local current density during the plating and stripping processes [46], leading to a homogeneous deposition of Li on Cu and an enhancement of the coulombic efficiency of the half-cell. Additionally, the BN_xGr_y layer clearly decreases the impedance of the Li/Cu half-cell, see Fig. 5i. In which, the pristine PP membrane exhibits an impedance of 95.5Ω , meanwhile the Gr/PP and BN/PP separators present impedances of 36Ω and 53.3Ω , respectively, being the lowest and highest impedance values among the modified separators.

The (de)lithiation reactions as well as the reduction of the interfacial resistance, could be the main reasons that promotes a remarkably stable plating and stripping behavior in the Li/Cu half-cells with the modified $\text{BN}_{50}\text{Gr}_{50}/\text{PP}$ separator.

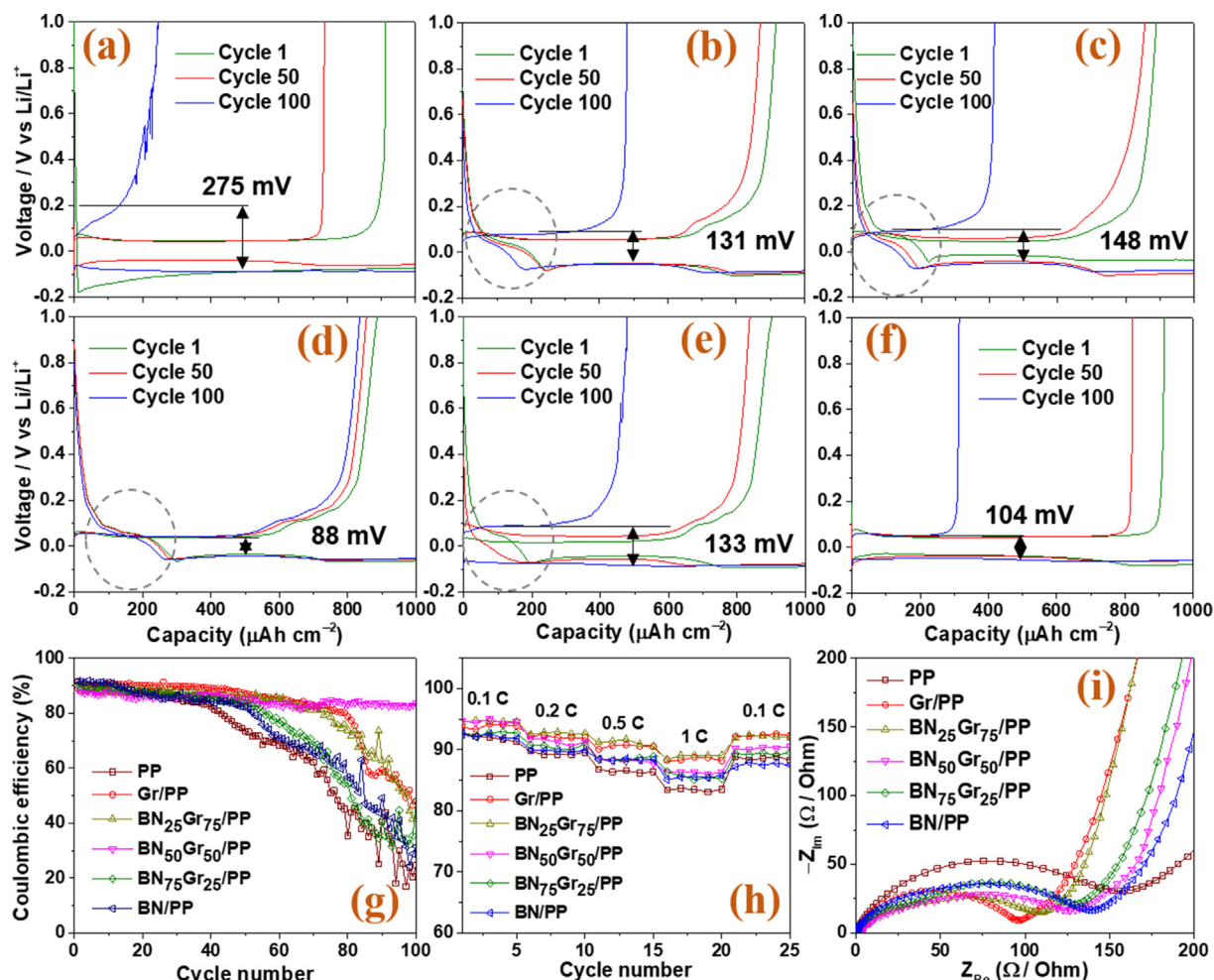


Fig. 5. Electrochemical plating and stripping profiles of (a) pristine PP membrane, (b) Gr nanoplatelets, (c) BN₂₅Gr₇₅, (d) BN₅₀Gr₅₀, (e) BN₇₅Gr₂₅ and (f) BN₁₀₀ on PP separator after 100 cycles at an applied current of 0.5 mA cm⁻². (g) cycle performance; (h) rate capability; and (i) EIS information.

Fig. 6 presents the images acquired from the cycled Li/Cu half-cells using the pristine and modified separators. After 100 cycles, the Cu foil presents a grayish color (Fig. 6a). In which, the Cu foil was entirely covered by Li dendrites that cannot be extracted via

the stripping process. The red arrows in Fig. 6b indicates some huge Li protuberances (>30 μm) that could easily penetrate the pristine PP separator and cause the short-circuiting of the cell [47]. These massive Li dendrites could continuously grow from

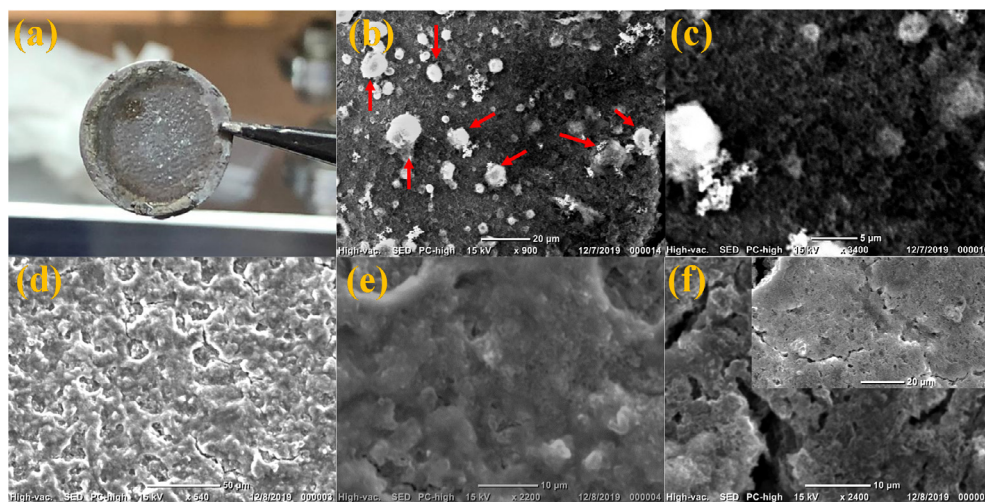


Fig. 6. Topographical analysis: (a) photo of the Cu foil after cycling; SEM image of the Li dendrites after 100 cycles using (b, c) the commercial PP membrane, and the modified separator with a layer of (d, e) BN₅₀Gr₅₀ and (f) BN.

the small dendrites exhibited on the Fig. 6c, until reaching their huge sizes [47]. In contrast, the utilization of the modified $\text{BN}_x\text{Gr}_y/\text{PP}$ separators suppresses the growth of those Li dendrites (Fig. 6d–f). Revealing that Li is deposited on a smooth way in comparison to the pristine PP membrane, leaving a flat surface without big Li protuberances or sharp Li dendrites, which leads to reduce the impedance of the cells as well as improve their performance as given in Fig. 5.

The impact of the modified $\text{BN}_x\text{Gr}_y/\text{PP}$ separators on the cycling performance of the Li/LFP full cell was evaluated via cyclability, rate capability and impedance studies by using galvanostatic (dis)charge cycling and EIS measurements. Fig. 7a–f present similar voltage profiles of the LFP cathode for cycle 1st, 100th, 500th and 1000th acquired at an applied C-rate of 1.0 C, with a voltage range from 2.5 to 4.2 V vs Li/Li^+ . Before, the cycle 500th, there is no significant difference between the average performance of the modified and pristine separators (Fig. 7g). But, from cycle 500th to 1000th, the modified separator presented a remarkably high and stable cycle performance. The modified $\text{BN}_{50}\text{Gr}_{50}/\text{PP}$ separator improves the Li/LFP battery performance, delivering a specific capacity of 114 mAh g^{-1} at 1.0C-rate with a high coulombic efficiency of 99% after 1000 cycles, and retaining 93% of its capacity, exhibiting the lowest electrode polarization (259 mV) after 1000

charge-discharge cycles. This value of charge capacity is 28% higher than that one delivered by the Li/LFP battery assembled with the bare PP separator, which loss capacity gradually. Fig. 7g shows clearly that the modification of the PP separator by coating a BN_xGr_y layer improve the long cycle performance of the Li/LFP full cell. Similarly, the rate capability study of the Li/LFP full cell (Fig. 7h) reveals that the modified separators boost its performance at low and high C-rate conditions and retains higher capacity (~10%) than the pristine PP separator, as result of a decrement of the interfacial resistance between the LFP electrode and the electrolyte, see Fig. 7i. As shown in the Nyquist plots, the pristine PP membrane displays the highest impedance with a value of 158Ω , meanwhile the Gr/PP and BN/PP separators present impedances of 96Ω and 139Ω , respectively.

The remarkably high performance of the modified PP separators, mainly the $\text{BN}_{50}\text{Gr}_{50}/\text{PP}$, is attributed to synergistic effect between physico-chemical properties of Gr and BN at the BN_xGr_y layer. Meanwhile the Gr nanoplatelets suppresses the applied current density during the plating and stripping processes via the (de)lithiation reactions, the BN sheets provide a high mechanical stability (Young modulus of 80 GPa and similar hardness of ~90 GPa) [38] and thermal conductivity ($80 \text{ W m}^{-1} \text{ K}^{-1}$) [41] (see Fig. S3). These suppress the growth of sharp Li dendrites, and then,

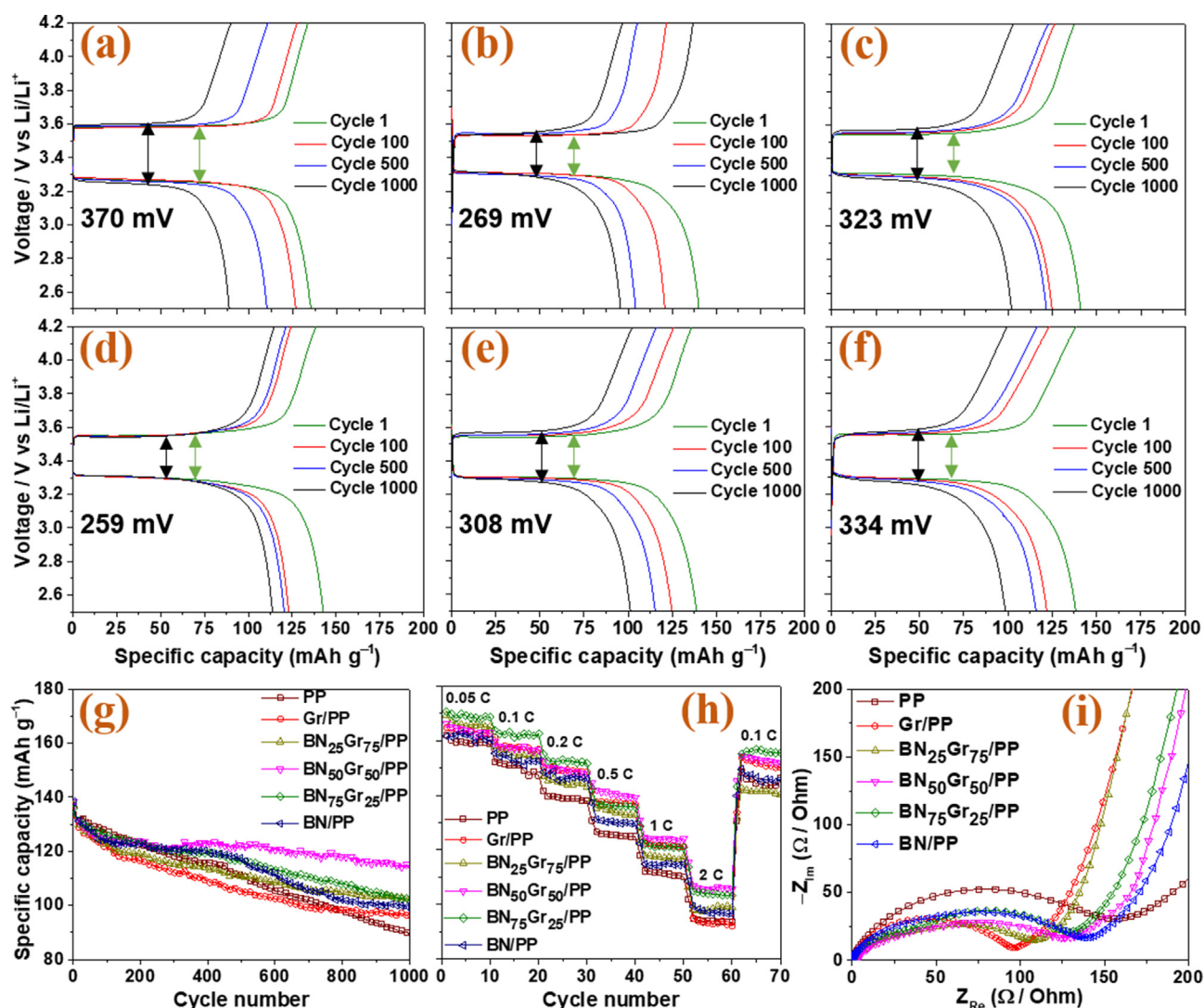


Fig. 7. Electrochemical properties of the Li/LFP full cells. Voltage profiles of (a) pristine PP membrane, and (b) Gr, (c) $\text{BN}_{25}\text{Gr}_{75}$, (d) $\text{BN}_{50}\text{Gr}_{50}$, (e) $\text{BN}_{75}\text{Gr}_{25}$ and (f) BN_{100} on PP separator after 1000 cycles at 1C-rate; (g) cycle performance; (h) rate capability; and (i) EIS information.

decreases of the impedance of the Li/Cu and Li/LFP cells, as result of an improvement on the electrochemical properties of the bare PP separator, which present a Young modulus of 0.25 GPa [48] and thermal conductivity $1 \text{ W m}^{-1} \text{ K}^{-1}$ [41].

4. Conclusion

In summary, via a facile and versatile one-pot procedure, ball-milling, thin BN_xGr_y nanosheets were prepared. This hybrid material demonstrated remarkably competence by improving the electrochemical properties of the commercial PP separator on Li/Cu half-cell and Li/LFP full cell configurations. The superiority of the modified $\text{BN}_{50}\text{Gr}_{50}/\text{PP}$ may be ascribed to synergistic effect between physico-chemical properties of Gr and BN that provide good structure stability and cycling performance with high rate capability, due to a high heat dissipation and mechanical stability as well as a reduction of the local current density as result of the (de)lithiation processes at Gr, leading to a smooth Li deposition instead of the growth of sharp Li dendrites.

CRedit authorship contribution statement

Jassiel R. Rodriguez: Conceptualization, Investigation. **Patrick J. Kim:** Conceptualization, Visualization. **Kyungho Kim:** Methodology. **Zhimin Qi:** Investigation. **Haiyan Wang:** Supervision. **Vilas G. Pol:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported in part by CONACYT-SENER project No. 274314 and U.S. Office of Naval Research (ONR, N00014-16-1-2465).

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcis.2020.09.009>.

References

- [1] C.P. Grey, J.M. Tarascon, Sustainability and in situ monitoring in battery development, *Nat. Mater.* 16 (2016) 45–56, <https://doi.org/10.1038/nmat4777>.
- [2] Y. Liu, Y. Zhu, Y. Cui, Challenges and opportunities towards fast-charging battery materials, *Nat. Energy* 4 (2019) 540–550, <https://doi.org/10.1038/s41560-019-0405-3>.
- [3] Z. Yang, J. Zhang, M.C.W. Kintner-Meyer, X. Lu, D. Choi, J.P. Lemmon, J. Liu, Electrochemical energy storage for green grid, *Chem. Rev.* 111 (2011) 3577–3613, <https://doi.org/10.1021/cr100290v>.
- [4] M.R. Palacin, Recent advances in rechargeable battery materials: a chemist's perspective, *Chem. Soc. Rev.* 38 (2009) 2565–2575, <https://doi.org/10.1039/b820555h>.
- [5] H.Y. Lee, J.K. Baek, S.M. Lee, H.K. Park, K.Y. Lee, M.H. Kim, Effect of carbon coating on elevated temperature performance of graphite as lithium-ion battery anode material, *J. Power Sources* 128 (2004) 61–66, <https://doi.org/10.1016/j.jpowsour.2003.09.051>.
- [6] R.A. Adams, A.N. Mistry, P.P. Mukherjee, V.G. Pol, Materials by design: tailored morphology and structures of carbon anodes for enhanced battery safety, *ACS Appl. Mater. Interfaces* 11 (2019) 13334–13342, <https://doi.org/10.1021/acsami.9b02921>.
- [7] M.N. Obrovac, V.L. Chevrier, Alloy negative electrodes for Li-ion batteries, *Chem. Rev.* 114 (2014) 11444–11502, <https://doi.org/10.1021/cr500207g>.
- [8] N. Yabuuchi, K. Kubota, M. Dahbi, S. Komaba, Research development on sodium-ion batteries, *Chem. Rev.* 114 (2014) 11636–11682, <https://doi.org/10.1021/cr500192f>.
- [9] W. Xu, J. Wang, F. Ding, X. Chen, E. Nasybulin, Y. Zhang, J.G. Zhang, Lithium metal anodes for rechargeable batteries, *Energy Environ. Sci.* 7 (2014) 513–537, <https://doi.org/10.1039/c3ee40795k>.
- [10] J.R. Rodriguez, S.B. Aguirre, V.G. Pol, Role of operando microscopy techniques on the advancement of sustainable sodium-ion battery anodes, *J. Power Sources* 437 (2019), <https://doi.org/10.1016/j.jpowsour.2019.226851>.
- [11] J. Liu, Z. Bao, Y. Cui, E.J. Dufek, J.B. Goodenough, P. Khalifah, Q. Li, B.Y. Liaw, P. Liu, A. Manthiram, Y.S. Meng, V.R. Subramanian, M.F. Toney, V.V. Viswanathan, M.S. Whittingham, J. Xiao, W. Xu, J. Yang, X.Q. Yang, J.G. Zhang, Pathways for practical high-energy long-cycling lithium metal batteries, *Nat. Energy* 4 (2019) 180–186, <https://doi.org/10.1038/s41560-019-0338-x>.
- [12] P.J. Kim, K. Kim, V.G. Pol, Uniform metal-ion flux through interface-modified membrane for highly stable metal batteries, *Electrochim. Acta* 283 (2018) 517–527, <https://doi.org/10.1016/j.electacta.2018.06.177>.
- [13] W. Mei, H. Chen, J. Sun, Q. Wang, The effect of electrode design parameters on battery performance and optimization of electrode thickness based on the electrochemical-thermal coupling model, *Sustain. Energy Fuels* 3 (2019) 148–165, <https://doi.org/10.1039/c8se00503f>.
- [14] Y. Zhu, X. He, Y. Mo, Strategies based on nitride materials chemistry to stabilize Li metal anode, *Adv. Sci.* 4 (2017) 1–11, <https://doi.org/10.1002/advs.201600517>.
- [15] G. Li, Z. Liu, Q. Huang, Y. Gao, M. Regula, D. Wang, L. Chen, D. Wang, Polyethylenimine sponge hosts by way of electrokinetic effects, *Nat. Energy* 3 (2018) 1076–1083, <https://doi.org/10.1038/s41560-018-0276-z>.
- [16] S. Choudhury, R. Mangal, A. Agrawal, L.A. Archer, A highly reversible room-temperature lithium metal battery based on crosslinked hairy nanoparticles, *Nat. Commun.* 6 (2015) 1–9, <https://doi.org/10.1038/ncomms10101>.
- [17] L. Chen, X. Fan, X. Ji, J. Chen, S. Hou, C. Wang, High-energy Li metal battery with lithiated host, *Joule* 3 (2019) 732–744, <https://doi.org/10.1016/j.joule.2018.11.025>.
- [18] J. Wang, W. Huang, A. Pei, Y. Li, F. Shi, X. Yu, Y. Cui, Improving cyclability of Li metal batteries at elevated temperatures and its origin revealed by cryo-electron microscopy, *Nat. Energy* 4 (2019) 664–670, <https://doi.org/10.1038/s41560-019-0413-3>.
- [19] H. Ye, S. Xin, Y.X. Yin, J.Y. Li, Y.G. Guo, L.J. Wan, Stable Li plating/stripping electrochemistry realized by a hybrid Li reservoir in spherical carbon granules with 3D conducting skeletons, *J. Am. Chem. Soc.* 139 (2017) 5916–5922, <https://doi.org/10.1021/jacs.7b01763>.
- [20] H.N. Umh, J. Park, J. Yeo, S. Jung, I. Nam, J. Yi, Lithium metal anode on a copper dendritic superstructure, *Electrochem. Commun.* 99 (2019) 27–31, <https://doi.org/10.1016/j.elecom.2018.12.015>.
- [21] N. Li, W. Wei, K. Xie, J. Tan, L. Zhang, X. Luo, K. Yuan, Q. Song, H. Li, C. Shen, E.M. Ryan, L. Liu, B. Wei, Suppressing dendritic lithium formation using porous media in lithium metal-based batteries, *Nano Lett.* 18 (2018) 2067–2073, <https://doi.org/10.1021/acs.nanolett.8b00183>.
- [22] X. Cheng, T. Hou, R. Zhang, H. Peng, C. Zhao, Dendrite-free lithium deposition induced by uniformly distributed lithium ions for efficient lithium metal batteries, *Adv. Mater.* 2 (2016) 2888–2895, <https://doi.org/10.1002/adma.201506124>.
- [23] X.X. Zeng, Y.X. Yin, N.W. Li, W.C. Du, Y.G. Guo, L.J. Wan, Reshaping lithium plating/stripping behavior via bifunctional polymer electrolyte for room-temperature solid Li metal batteries, *J. Am. Chem. Soc.* 138 (2016) 15825–15828, <https://doi.org/10.1021/jacs.6b10088>.
- [24] P. Shi, L. Zhang, H. Xiang, X. Liang, Y. Sun, W. Xu, Lithium difluorophosphate as a dendrite-suppressing additive for lithium metal batteries, *ACS Appl. Mater. Interfaces* 10 (2018) 22201–22209, <https://doi.org/10.1021/acsami.8b05185>.
- [25] S. Choudhury, A. Agrawal, S. Wei, E. Jeng, L.A. Archer, Hybrid hairy nanoparticle electrolytes stabilizing lithium metal batteries, *Chem. Mater.* 28 (2016) 2147–2157, <https://doi.org/10.1021/acs.chemmater.6b00029>.
- [26] K. Park, J.B. Goodenough, Dendrite-suppressed lithium plating from a liquid electrolyte via wetting of Li₃N, *Adv. Energy Mater.* 7 (2017) 1–7, <https://doi.org/10.1002/aenm.201700732>.
- [27] C. Zhang, Q. Lan, Y. Liu, J. Wu, H. Shao, H. Zhan, Y. Yang, A dual-layered artificial solid electrolyte interphase formed by controlled electrochemical reduction of LiTFSI/DME-LiNO₃ for dendrite-free lithium metal anode, *Electrochim. Acta* 306 (2019) 407–419, <https://doi.org/10.1016/j.electacta.2019.03.162>.
- [28] B. Zhu, Y. Jin, X. Hu, Q. Zheng, S. Zhang, Q. Wang, J. Zhu, Poly(dimethylsiloxane) thin film as a stable interfacial layer for high-performance lithium-metal battery anodes, *Adv. Mater.* 29 (2017) 2–7, <https://doi.org/10.1002/adma.201603755>.
- [29] Z. Zhu, L.L. Lu, Y. Yin, J. Shao, B. Shen, H. Bin Yao, High rate and stable solid-state lithium metal batteries enabled by electronic and ionic mixed conducting network interlayers, *ACS Appl. Mater. Interfaces* 11 (2019) 16578–16585, <https://doi.org/10.1021/acsami.9b02184>.
- [30] M.H. Ryou, D.J. Lee, J.N. Lee, Y.M. Lee, J.K. Park, J.W. Choi, Excellent cycle life of lithium-metal anodes in lithium-ion batteries with mussel-inspired polydopamine-coated separators, *Adv. Energy Mater.* 2 (2012) 645–650, <https://doi.org/10.1002/aenm.201100687>.
- [31] P.J.H. Kim, V.G. Pol, Surface functionalization of a conventional polypropylene separator with an aluminum nitride layer toward ultrastable and high-rate lithium metal anodes, *ACS Appl. Mater. Interfaces* 11 (2019) 3917–3924, <https://doi.org/10.1021/acsami.8b18660>.
- [32] P.J. Kim, V.G. Pol, High performance lithium metal batteries enabled by surface tailoring of polypropylene separator with a polydopamine/graphene layer, *Adv. Energy Mater.* 8 (2018) 1–8, <https://doi.org/10.1002/aenm.201802665>.

- [33] M. Waqas, S. Ali, W. Lv, D. Chen, B. Boateng, W. He, High-performance PE-BN/PVDF-HFP bilayer separator for lithium-ion batteries, *Adv. Mater. Interfaces* 6 (2019) 1–10, <https://doi.org/10.1002/admi.201801330>.
- [34] T. Nestler, R. Schmid, W. Münchgesang, V. Bazhenov, J. Schilm, T. Leisegang, D. C. Meyer, Separators – technology review: ceramic based separators for secondary batteries, 2014. doi:10.1063/1.4878486.
- [35] G. Yoon, S. Moon, G. Ceder, K. Kang, Deposition and stripping behavior of lithium metal in electrochemical system: continuum mechanics study, *Chem. Mater.* 30 (2018) 6769–6776, <https://doi.org/10.1021/acs.chemmater.8b02623>.
- [36] Y. Zhang, C. Pan, Measurements of mechanical properties and number of layers of graphene from nano-indentation, *Diam. Relat. Mater.* 24 (2012) 1–5, <https://doi.org/10.1016/j.diamond.2012.01.033>.
- [37] A. Falin, Q. Cai, E.J.G. Santos, D. Scullion, D. Qian, R. Zhang, Z. Yang, S. Huang, K. Watanabe, T. Taniguchi, M.R. Barnett, Y. Chen, R.S. Ruoff, L.H. Li, Mechanical properties of atomically thin boron nitride and the role of interlayer interactions, *Nat. Commun.* 8 (2017) 1–9, <https://doi.org/10.1038/ncomms15815>.
- [38] Q. Cai, D. Scullion, W. Gan, A. Falin, S. Zhang, K. Watanabe, T. Taniguchi, Y. Chen, E.J.G. Santos, L.H. Li, High thermal conductivity of high-quality monolayer boron nitride and its thermal expansion, *Sci. Adv.* 5 (2019) 1–9, <https://doi.org/10.1126/sciadv.aav0129>.
- [39] J.Y. Kim, D.O. Shin, K.M. Kim, J. Oh, J. Kim, S.H. Kang, M.J. Lee, Y.G. Lee, Graphene oxide induced surface modification for functional separators in lithium secondary batteries, *Sci. Rep.* 9 (2019) 1–7, <https://doi.org/10.1038/s41598-019-39237-8>.
- [40] A.A. Assegie, C.C. Chung, M.C. Tsai, W.N. Su, C.W. Chen, B.J. Hwang, Multilayer-graphene-stabilized lithium deposition for anode-free lithium-metal batteries, *Nanoscale* 11 (2019) 2710–2720, <https://doi.org/10.1039/c8nr06980h>.
- [41] W. Luo, L. Zhou, K. Fu, Z. Yang, J. Wan, M. Manno, Y. Yao, H. Zhu, B. Yang, L. Hu, A thermally conductive separator for stable Li metal anodes, 2015. doi:10.1021/acs.nanolett.5b02432.
- [42] X. Li, J. Liu, K. Ding, X. Zhao, S. Li, W. Zhou, B. Liang, Temperature dependence of Raman-active in-plane E_{2g} phonons in layered graphene and h-BN flakes, *Nanoscale Res. Lett.* 13 (2018), <https://doi.org/10.1186/s11671-018-2444-2>.
- [43] S. Yuan, B. Toury, C. Journet, A. Brioude, Synthesis of hexagonal boron nitride graphene-like few layers, *Nanoscale* 6 (2014) 7838–7841, <https://doi.org/10.1039/c4nr01017e>.
- [44] N. Kumar, J.R. Rodriguez, V.G. Pol, A. Sen, Applied surface science facile synthesis of 2D graphene oxide sheet enveloping ultra fine 1D LiMn₂O₄ as interconnected framework to enhance cathodic property for Li-ion battery, *Appl. Surf. Sci.* 463 (2019) 132–140, <https://doi.org/10.1016/j.apsusc.2018.08.210>.
- [45] H.H. Nersisyan, T.H. Lee, K.H. Lee, Y.S. An, J.S. Lee, J.H. Lee, Few-atomic-layer boron nitride nanosheets synthesized in solid thermal waves, *RSC Adv.* 5 (2015) 8579–8584, <https://doi.org/10.1039/c4ra10907d>.
- [46] Q. Zhao, X. Hao, S. Su, J. Ma, Y. Hu, Y. Liu, F. Kang, Y.B. He, Expanded-graphite embedded in lithium metal as dendrite-free anode of lithium metal batteries, *J. Mater. Chem. A* 7 (2019) 15871–15879, <https://doi.org/10.1039/c9ta04240g>.
- [47] X. Guan, A. Wang, S. Liu, G. Li, F. Liang, Y.W. Yang, X. Liu, J. Luo, Controlling nucleation in lithium metal anodes, *Small* 14 (2018) 1–21, <https://doi.org/10.1002/smll.201801423>.
- [48] S. Kalnaus, Y. Wang, J.A. Turner, Mechanical behavior and failure mechanisms of Li-ion battery separators, *J. Power Sources* 348 (2017) 255–263, <https://doi.org/10.1016/j.jpowsour.2017.03.003>.